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Course:

Parallel and Distributed Computing

Section:

BCS-6(B)

Software Installation Guide and Technical Overview For Windows

APACHE AIRFLOW (Workflow Management Tool)

1. Background and History

Apache Airflow was first created in 2014 by Maxime Beauchemin while working at Airbnb.

Problem: Airbnb had a huge amount of data and many tasks that needed to run in a proper sequence every day. Managing these tasks manually or with basic scripts caused mistakes and confusion.

Solution: To solve this issue, Airflow was developed. It allows developers to create workflows using Python code instead of doing everything manually.

Growth: In 2016, Airflow became part of the Apache Software Foundation, and later it became one of its major projects. Now it is widely used in data engineering.

2. Key Features

DAGs (Directed Acyclic Graphs): These define the workflow. They show all tasks and how they depend on each other.

Scheduler: This works like a controller. It runs tasks at the correct time automatically.

User Interface (UI): Airflow provides a web dashboard where users can monitor tasks, check errors, and restart processes.

Scalability: It can work on a small system as well as large cloud environments with many tasks.

3. Development and Deployment

Development: Airflow is built using Python. It uses Flask for the web interface and SQLAlchemy to communicate with the database.

Deployment: In real-world use, Airflow is usually run inside Docker containers. This helps ensure the same setup works on every system.

4. Parallel or Distributed System

Apache Airflow is a distributed system. It works using different components such as:
Web Server
Scheduler
Database (PostgreSQL)
Message Broker (Redis)
These components run separately but work together. It also supports parallel processing, which means multiple tasks can run at the same time.

DOCKER DESKTOP (System Environment Tool)

1. What is Docker Desktop and Why is it Used?

Apache Airflow needs several tools (like Python, database, Redis), which are difficult to install manually on Windows.

Docker's role: Docker groups all required software into containers. Each container has everything needed to run properly.

Why needed on Windows: Airflow is mainly designed for Linux. Docker Desktop allows Windows users to run Linux-based applications easily using WSL 2 (Windows Subsystem for Linux).

2. Background and History

Docker was introduced in 2013 by Solomon Hykes. Before Docker, developers used virtual machines, which were slow and used a lot of system resources. Docker improved this by introducing containers, which are faster and lightweight.

3. Key Features

Containerization: Keeps software separate from the main system, avoiding conflicts.

Docker Compose: Allows running multiple containers together using a single file. This is useful for Airflow because it needs multiple services.

Graphical Dashboard: Provides a visual interface to check whether containers are running or stopped.

WSL 2: To support Linux environment on Windows

4. Development and Deployment

Development: Docker’s core system is written in Go (Golang). Its interface uses React and Electron.
Deployment: It is installed as an application on your computer and acts as a platform to run other software like Airflow.

How Apache Airflow Works

Apache Airflow works by organizing tasks into a structured workflow and running them step by step.
Step 1: Creating a DAG
First, the user writes a workflow using Python code, called a DAG (Directed Acyclic Graph). This DAG defines: What tasks need to be done and In what order they should run.
Step 2: Scheduling the Workflow
The Scheduler reads the DAG file and checks the defined schedule (for example: daily, hourly). It decides when to start the workflow.
Step 3: Task Execution
Each task in the DAG is sent to workers (or executor) for execution. Tasks run based on their dependencies, meaning: A task will only run when its previous task is completed.
Step 4: Monitoring through UI
Airflow provides a web interface where users can:
See running and completed tasks
Check errors or logs
Restart failed tasks
Step 5: Storing Information
All task information (status, logs, history) is stored in a database.

Summary:
We use Docker as a base platform where everything runs. Inside Docker, we run Apache Airflow, which manages and schedules tasks. Since Airflow is a distributed

system, it uses different containers for its components like database, scheduler, and web interface. Using Python, we define what tasks should run and when they should execute.

Installation steps

Download Docker Desktop: Go to the official Docker documentation website and download the installer for Windows.

Run the Installer: Open the downloaded Docker Desktop Installer.exe file to start the installation process.

Configuration: In the configuration window, ensure the option "Use WSL 2 instead of Hyper-V" is selected for better performance.

Wait for Installation: The installer will unpack the necessary files. Once finished, click "Close and restart" to reboot your computer.

Accept Terms: After restarting, open Docker Desktop and Accept the Service Agreement.

Update WSL: Open Windows PowerShell as an administrator and run the command wsl --update to ensure the Windows Subsystem for Linux is up to date.

Set Up Apache Airflow:

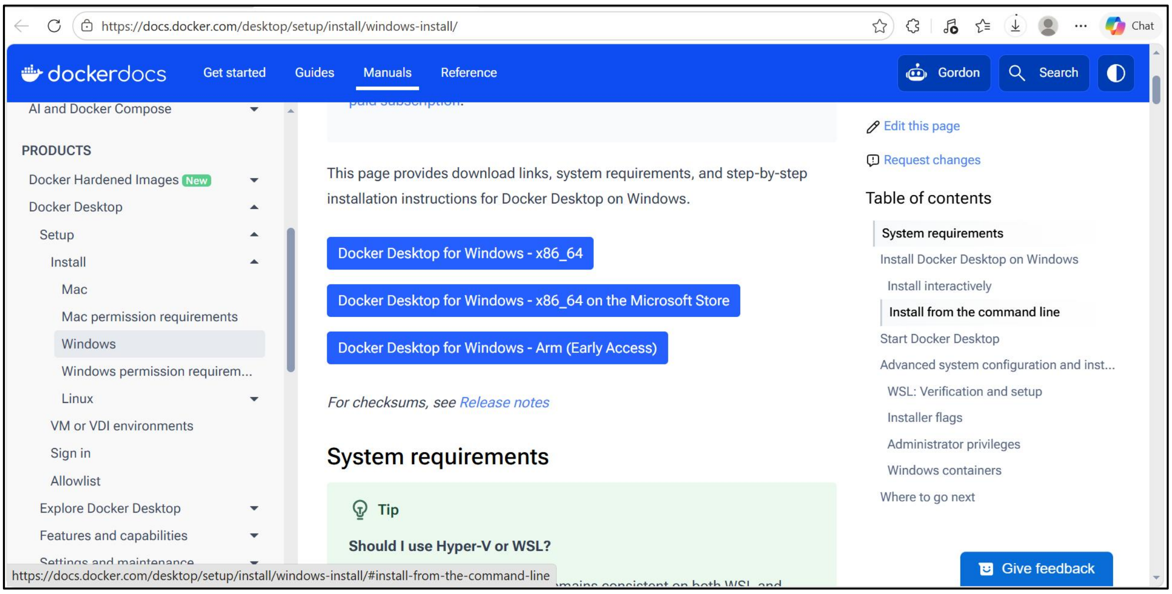
Open your project folder in VS Code.

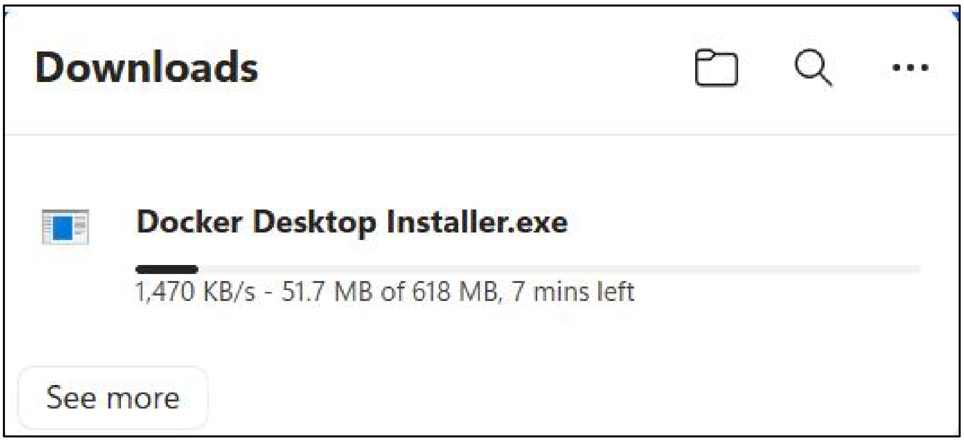
In the terminal, use the command docker-compose up -d to pull the required images (Airflow, Postgres, Redis) and start the containers.

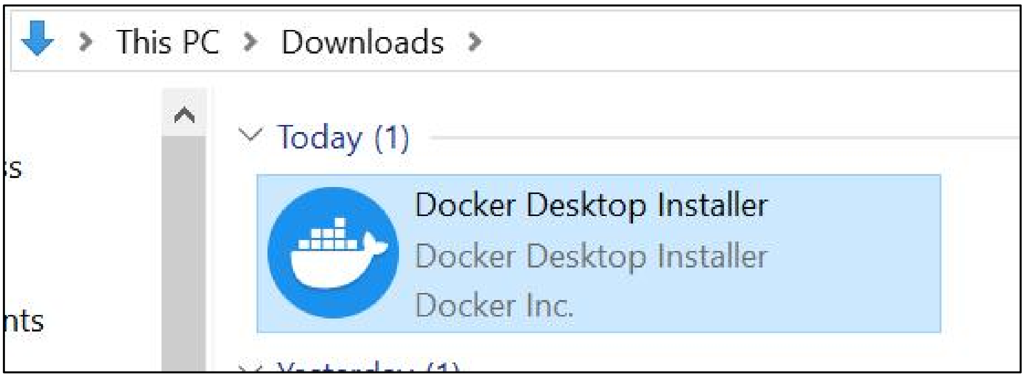
Create an admin user by running the airflow users create command in the terminal.

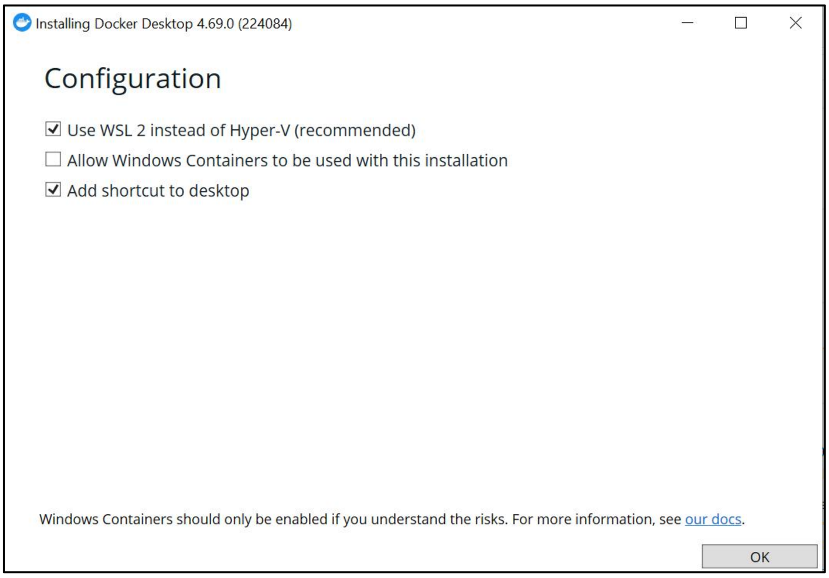
Access the Dashboard: Open your web browser and go to localhost:8080. Log in with the admin credentials you just created to access the Airflow dashboard.

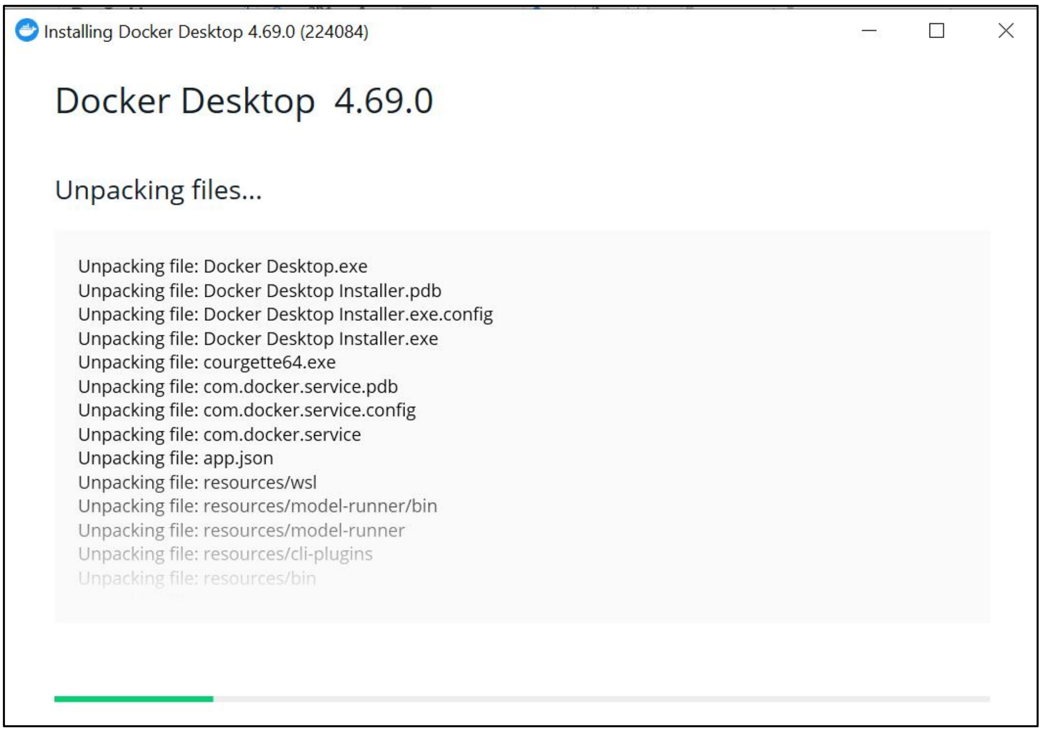
Screenshots:

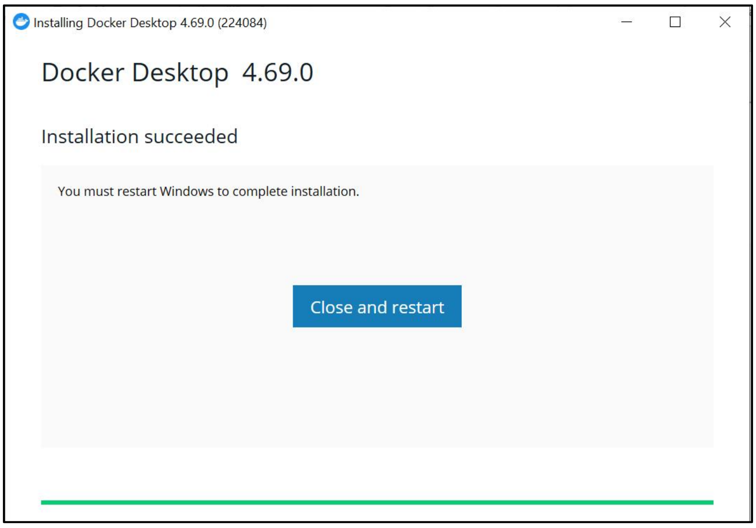


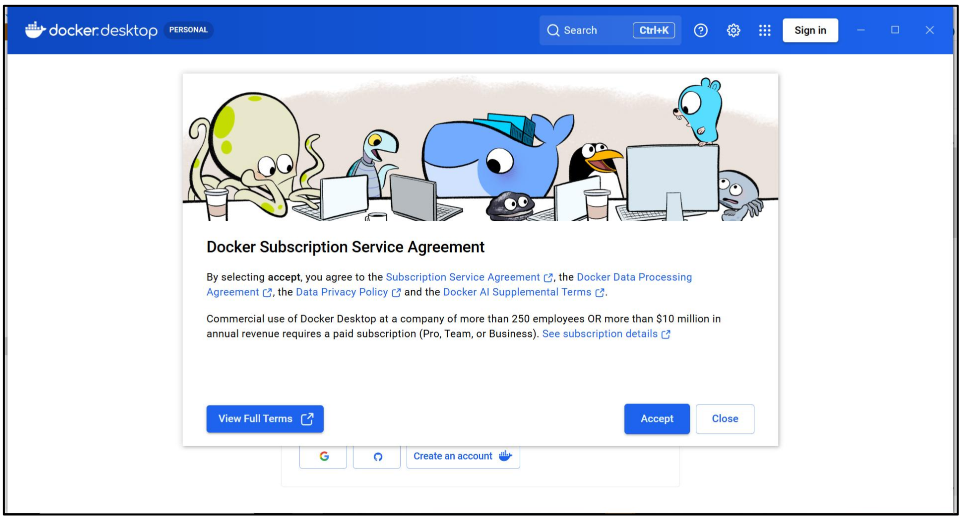


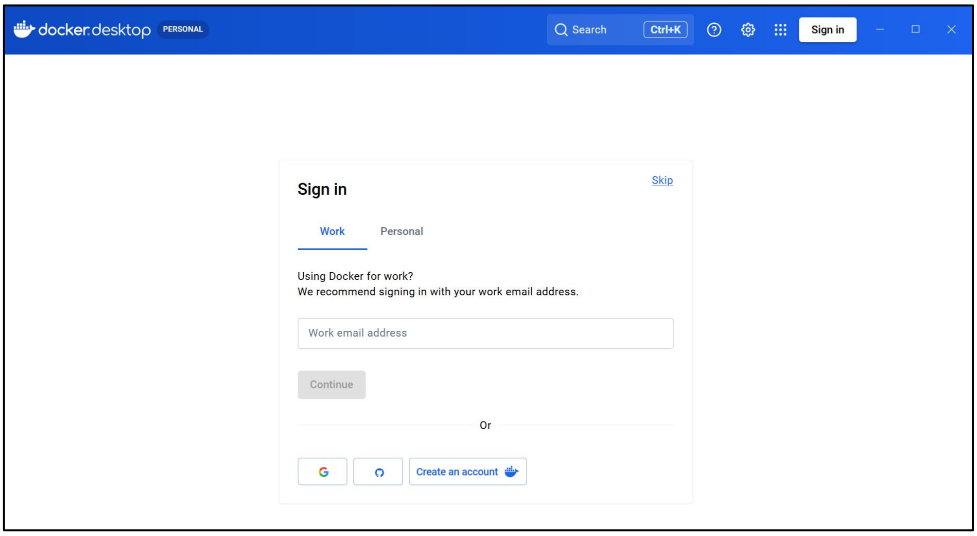


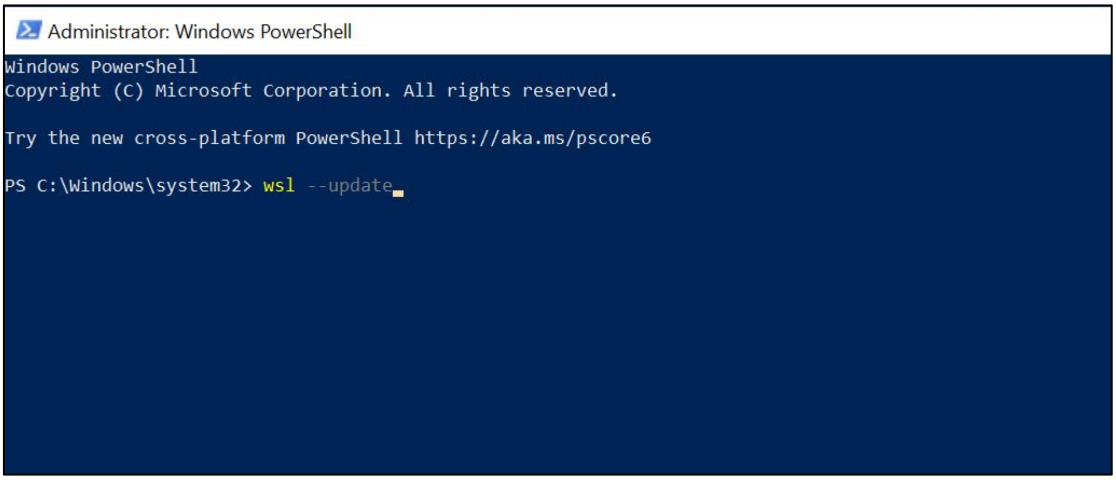

Installing Docker Desktop 4.69.0 (224084)
Configuration
Use WSL 2 instead of Hyper-V (recommended)
Allow Windows Containers to be used with this installation
Add shortcut to desktop
Windows Containers should only be enabled if you understand the risks. For more information, see our docs.
OK

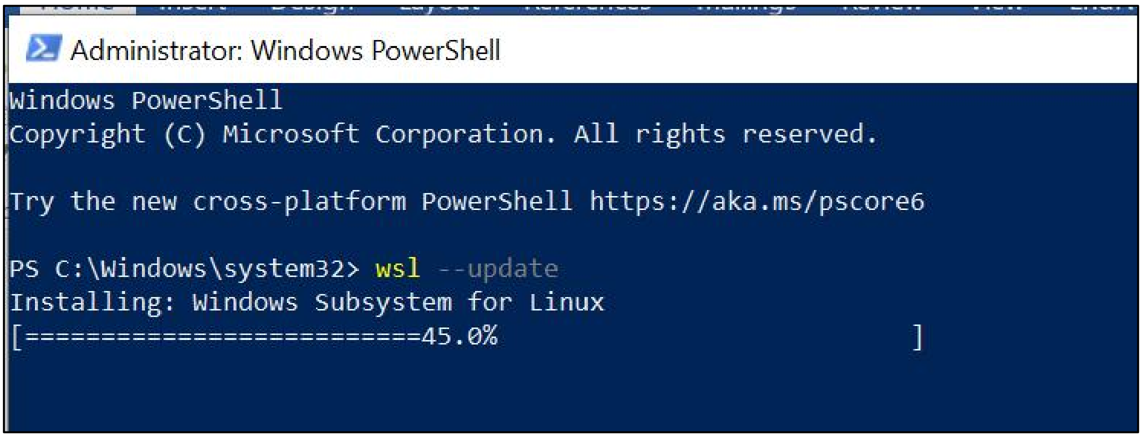

Installing Docker Desktop 4.69.0 (224084)
Docker Desktop 4.69.0
Unpacking files...
Unpacking file: Docker Desktop.exe
Unpacking file: Docker Desktop Installer.pdb
Unpacking file: Docker Desktop Installer.exe.config
Unpacking file: Docker Desktop Installer.exe
Unpacking file: courgette64.exe
Unpacking file: com.docker.service.pdb
Unpacking file: com.docker.service.config
Unpacking file: com.docker.service
Unpacking file: app.json
Unpacking file: resources/wsl
Unpacking file: resources/model-runner/bin
Unpacking file: resources/model-runner
Unpacking file: resources/cli-plugins
Unpacking file: resources/bin

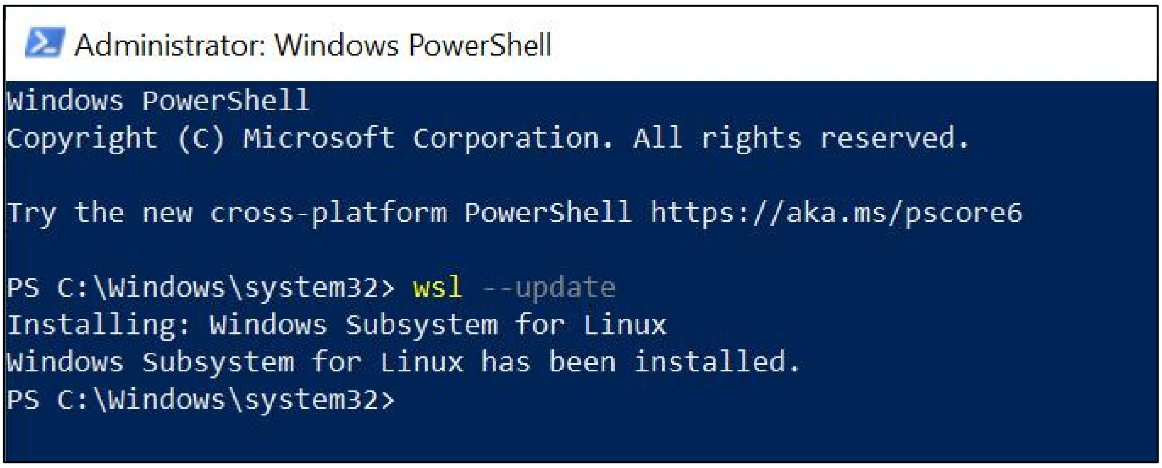

Installing Docker Desktop 4.69.0 (224084)
Docker Desktop 4.69.0
Installation succeeded
You must restart Windows to complete installation.
Close and restart

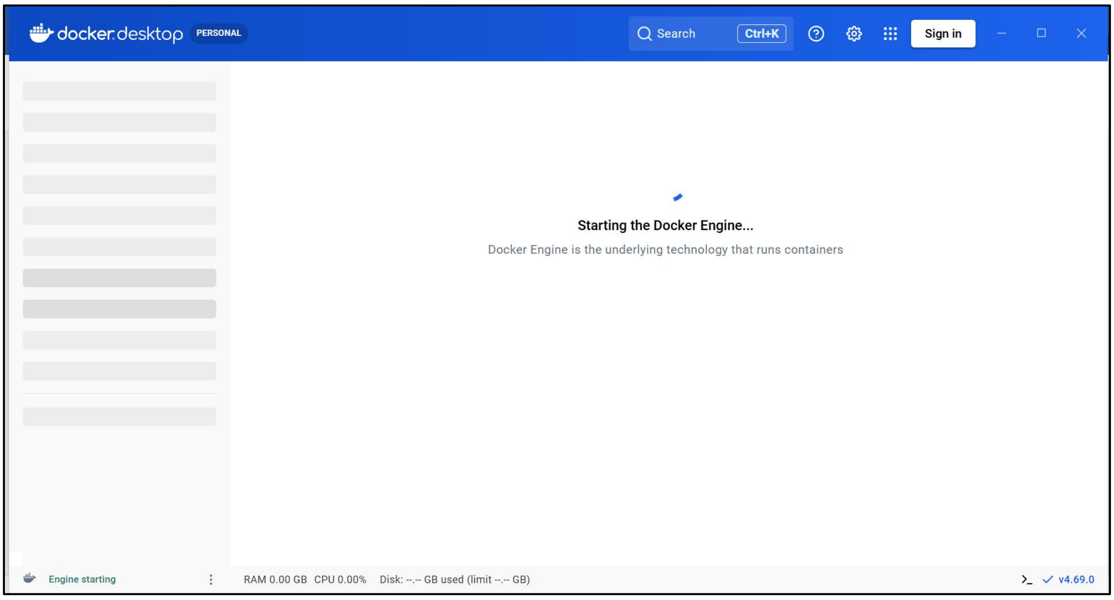

docker desktop PERSONAL
Search
Ctrl+K
Sign in
Docker Subscription Service Agreement
By selecting accept, you agree to the Subscription Service Agreement, the Docker Data Processing Agreement, the Data Privacy Policy and the Docker AI Supplemental Terms.
Commercial use of Docker Desktop at a company of more than 250 employees OR more than $10 million in annual revenue requires a paid subscription (Pro, Team, or Business). See subscription details
View Full Terms
Accept
Close
Create an account

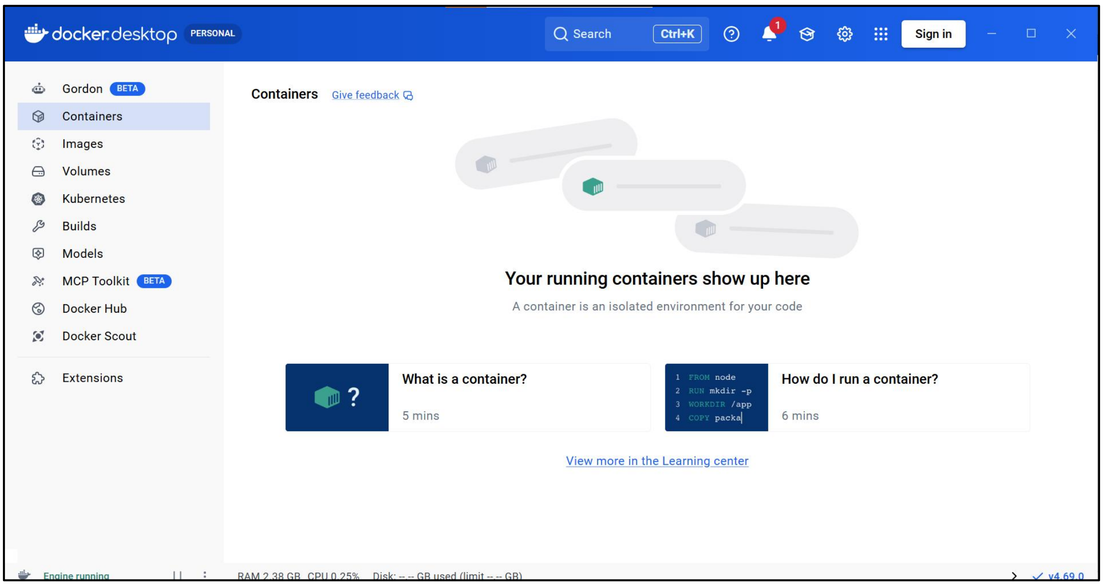

docker desktop PERSONAL
Search
Ctrl+K
Sign in
Sign in
Skip
Work
Personal
Using Docker for work?
We recommend signing in with your work email address.
Work email address
Continue
Or
Create an account


Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Try the new cross-platform PowerShell https://aka.ms/pscore6
PS C:\Windows\system32> wsl --update

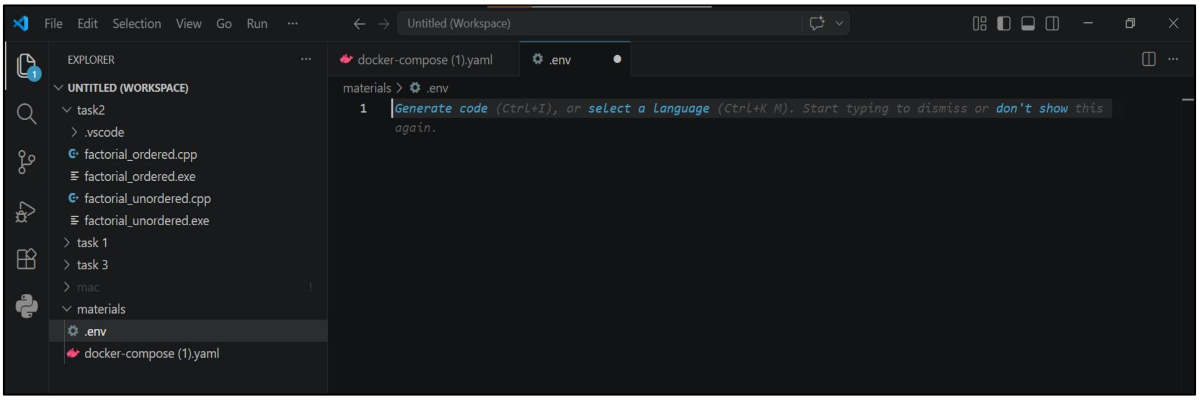

Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Try the new cross-platform PowerShell https://aka.ms/pscore6
PS C:\Windows\system32> wsl --update
Installing: Windows Subsystem for Linux
[=========================45.0%   ]

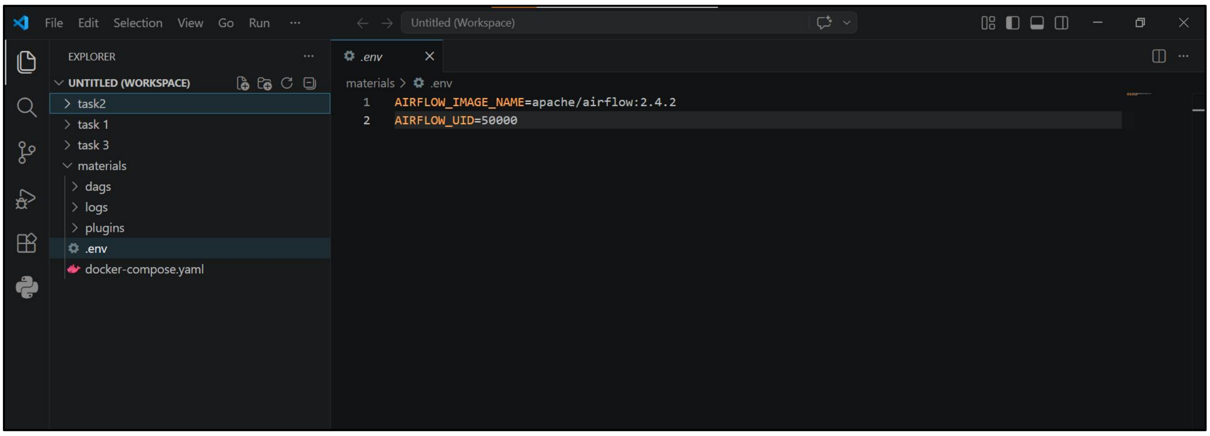

Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Try the new cross-platform PowerShell https://aka.ms/pscore6
PS C:\Windows\system32> wsl --update
Installing: Windows Subsystem for Linux
Windows Subsystem for Linux has been installed.
PS C:\Windows\system32>

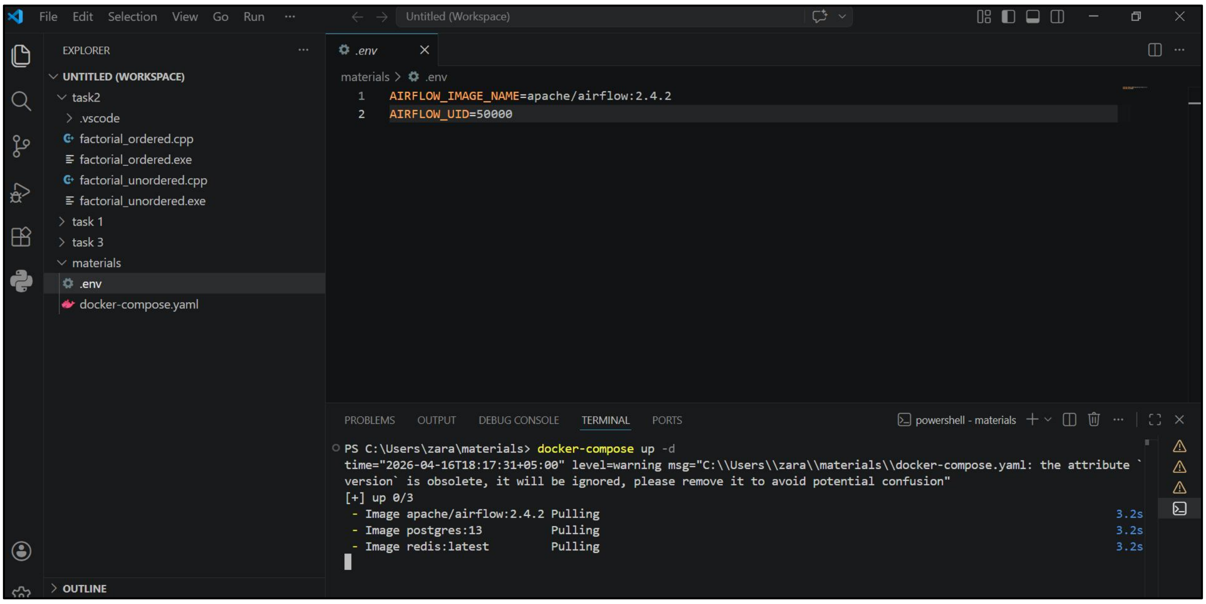

docker desktop PERSONAL
Search Ctrl+K
Sign in
Starting the Docker Engine...
Docker Engine is the underlying technology that runs containers
Engine starting
RAM 0.00 GB CPU 0.00% Disk: --.-- GB used (limit --.-- GB)
v4.69.0

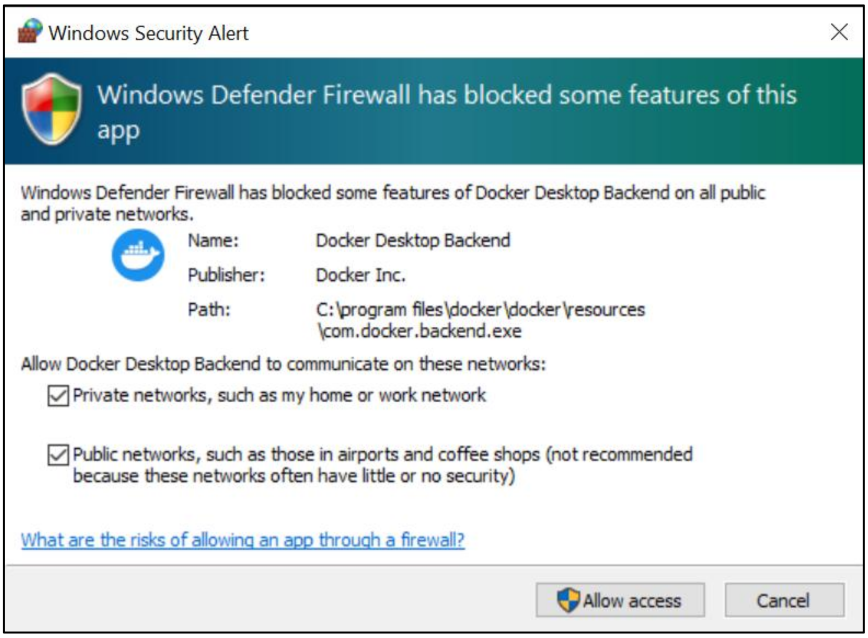

docker desktop PERSONAL
Search Ctrl+K
Sign in
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Containers Give feedback
Your running containers show up here
A container is an isolated environment for your code
What is a container?
5 mins
How do I run a container?
6 mins
View more in the Learning center

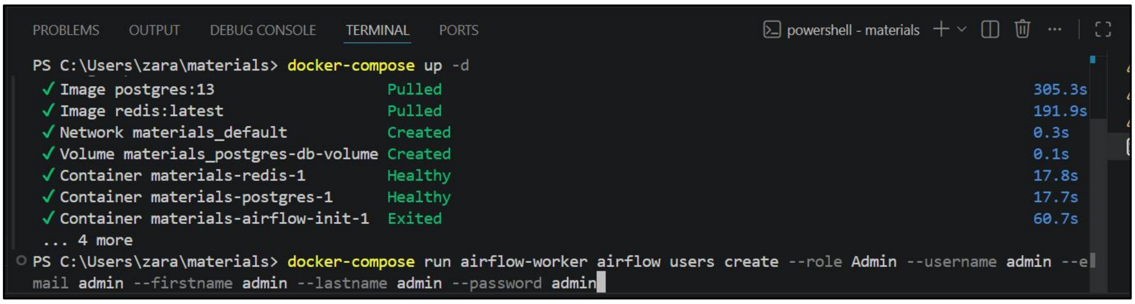

Downloads
docker-compose.yaml
Open file

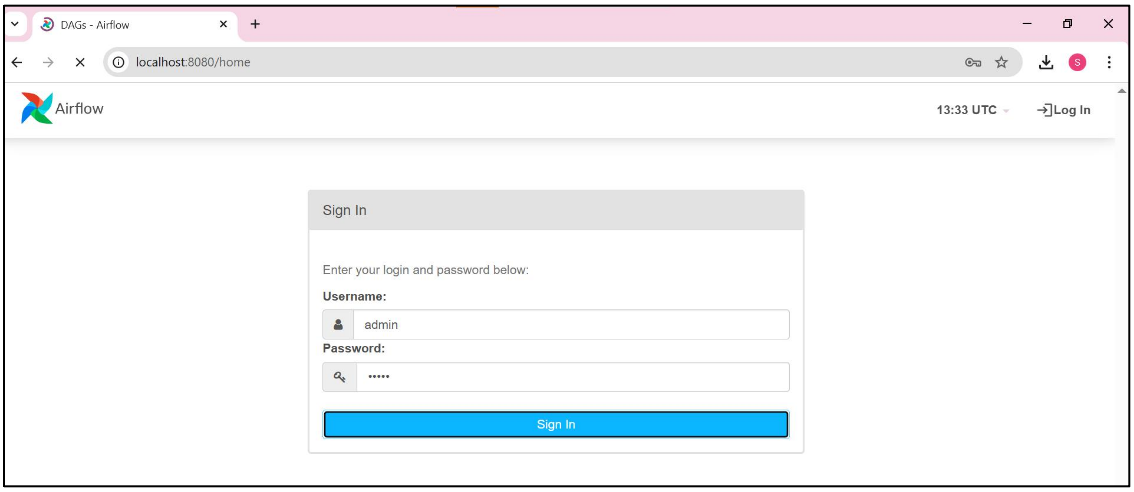

File Edit Selection View Go Run ···
Untitled (Workspace)
EXPLORER
UNTITLED (WORKSPACE)
task2
.vscode
factorial_ordered.cpp
factorial_ordered.exe
factorial_unordered.cpp
factorial_unordered.exe
task 1
task 3
mac
materials
.env
docker-compose (1).yaml
docker-compose (1).yaml
.env
materials > .env
1 Generate code (Ctrl+I), or select a language (Ctrl+K M). Start typing to dismiss or don't show this again.

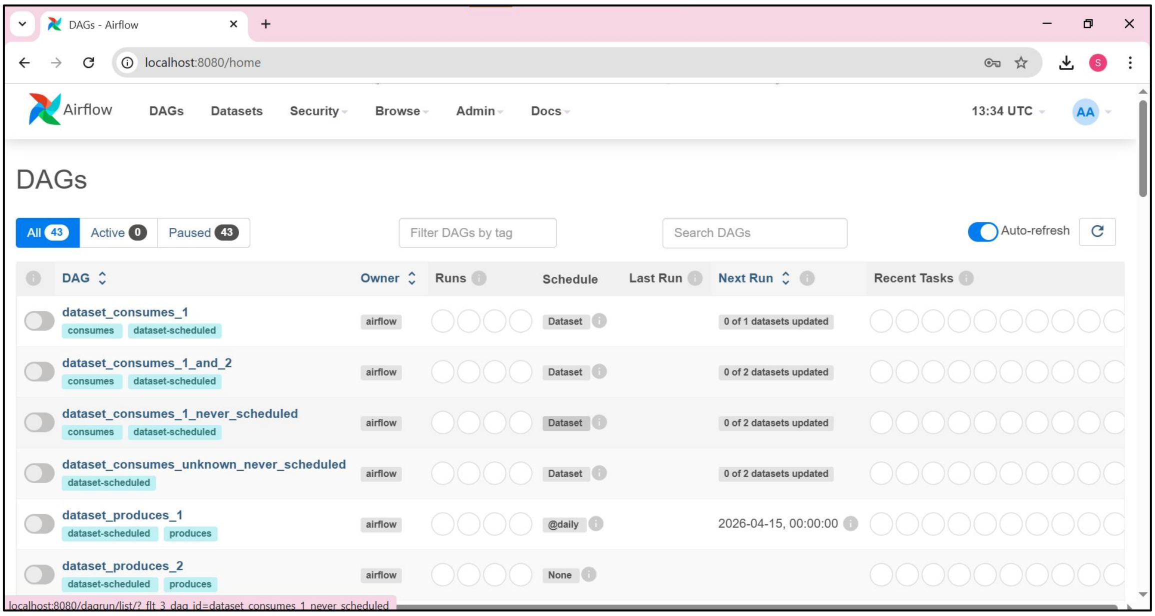

File Edit Selection View Go Run ···
Untitled (Workspace)
EXPLORER
UNTITLED (WORKSPACE)
task2
task 1
task 3
materials
dags
logs
plugins
.env
docker-compose.yaml
.env
materials > .env
1 AIRFLOW_IMAGE_NAME=apache/airflow:2.4.2
2 AIRFLOW_UID=50000


File Edit Selection View Go Run ···
Untitled (Workspace)
EXPLORER
UNTITLED (WORKSPACE)
task2
.vscode
factorial_ordered.cpp
factorial_ordered.exe
factorial_unordered.cpp
factorial_unordered.exe
task 1
task 3
materials
.env
docker-compose.yaml
OUTLINE
.env
materials > .env
1 AIRFLOW_IMAGE_NAME=apache/airflow:2.4.2
2 AIRFLOW_UID=50000
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell - materials
PS C:\Users\zara\materials> docker-compose up -d
time="2026-04-16T18:17:31+05:00" level=warning msg="C:\\Users\\zara\\materials\\docker-compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 0/3
- Image apache/airflow:2.4.2 Pulling 3.2s
- Image postgres:13 Pulling 3.2s
- Image redis:latest Pulling 3.2s


Windows Security Alert
Windows Defender Firewall has blocked some features of this app
Windows Defender Firewall has blocked some features of Docker Desktop Backend on all public and private networks.
Name: Docker Desktop Backend
Publisher: Docker Inc.
Path: C:\program files\docker\docker\resources \com.docker.backend.exe
Allow Docker Desktop Backend to communicate on these networks:
Private networks, such as my home or work network
Public networks, such as those in airports and coffee shops (not recommended because these networks often have little or no security)
What are the risks of allowing an app through a firewall?
Allow access
Cancel


PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell - materials
PS C:\Users\zara\materials> docker-compose up -d
✓ Image postgres:13 Pulled 305.3s
✓ Image redis:latest Pulled 191.9s
✓ Network materials_default Created 0.3s
✓ Volume materials_postgres-db-volume Created 0.1s
✓ Container materials-redis-1 Healthy 17.8s
✓ Container materials-postgres-1 Healthy 17.7s
✓ Container materials-airflow-init-1 Exited 60.7s
... 4 more
PS C:\Users\zara\materials> docker-compose run airflow-worker airflow users create --role Admin --username admin --email admin --firstname admin --lastname admin --password admin


DAGs - Airflow
localhost:8080/home
Airflow
13:33 UTC
Log In
Sign In
Enter your login and password below:
Username:
admin
Password:
•••••
Sign In


DAGs - Airflow
localhost:8080/home
Airflow
DAGs
Datasets
Security
Browse
Admin
Docs
13:34 UTC
AA
DAGs
All 43
Active 0
Paused 43
Filter DAGs by tag
Search DAGs
Auto-refresh
DAG
Owner
Runs
Schedule
Last Run
Next Run
Recent Tasks
dataset_consumes_1
consumes
dataset-scheduled
airflow
Dataset
0 of 1 datasets updated
dataset_consumes_1_and_2
consumes
dataset-scheduled
airflow
Dataset
0 of 2 datasets updated
dataset_consumes_1_never_scheduled
consumes
dataset-scheduled
airflow
Dataset
0 of 2 datasets updated
dataset_consumes_unknown_never_scheduled
dataset-scheduled
airflow
Dataset
0 of 2 datasets updated
dataset_produces_1
dataset-scheduled
produces
airflow
@daily
2026-04-15, 00:00:00
dataset_produces_2
dataset-scheduled
produces
airflow
None
localhost:8080/dagrun/list/?_flt_3_dag_id=dataset_consumes_1_never_scheduled

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